

Open Source for Regulated Manufacturing

GMP-Ready IIoT Data Logging with Apache IoTDB

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Slide 1: Open Source for Regulated Manufacturing: GMP-Ready IIoT Data Logging with Apache IoTDB

Welcome everyone. I'm Lukas Ott, Enterprise Architect, and today I want to show how open source can support scalable, inspectable, and GMP-ready IIoT data platforms.

The key message is simple: compliance does not come from a database alone.

It comes from the validated system around it: controlled ingestion, traceability, access control, retention, monitoring, testing, and lifecycle governance. [highlight]#This is the practical foundation for GMP-ready IIoT.#

The Pressure

Connected Manufacturing Meets Regulation



Slide 2: The Pressure +

Manufacturing is becoming more connected, more automated, and increasingly data-driven.

Machines, sensors, PLCs, vision systems, and edge devices generate high-frequency data. That data is no longer only useful for dashboards. In regulated manufacturing, machine data can become part of quality decisions, investigations, and inspection evidence.

That creates a tension: we want real-time data and scalable analytics, but we also need trustworthy records, controlled systems, and long-term inspection readiness.

The Shopfloor Is Becoming a Data Platform



Slide 3: The Shopfloor Is Becoming a Data Platform

The shopfloor is no longer a collection of isolated machines. It is becoming a distributed data platform.

But many shopfloor architectures were built machine by machine, site by site, and vendor by vendor. That often leads to fragmented semantics, inconsistent data models, proprietary interfaces, and technology stacks that are difficult to validate end to end.

So the question for today is: how can we build a scalable IIoT data foundation without losing control, traceability, and trust?

The Core Tension

- Manufacturing needs more real-time data.
- GMP requires trustworthy and inspectable records.
- AI and analytics increase the demand for structured, contextualized data.
- Legacy shopfloor architectures are often fragmented and opaque.

The future shopfloor needs data platforms that are both scalable and inspectable.

Slide 4: The Core Tension

This is the core tension of the talk.

On one side, manufacturing wants more connectivity, more automation, more analytics, and eventually AI-supported decision-making.

On the other side, GMP environments require evidence. The system must show where data came from, when it was captured, whether it was changed, who or what changed it, and whether it can be retrieved when needed.

Regulatory Pressure Is Expanding



Slide 5: Regulatory Pressure Is Expanding

The pressure is not only more data. It is more connected systems, more cloud, more cybersecurity expectations, and more AI-supported processes.

For regulated manufacturing, the question is no longer only whether a single record is correct. The question is whether the full computerized system lifecycle is controlled.

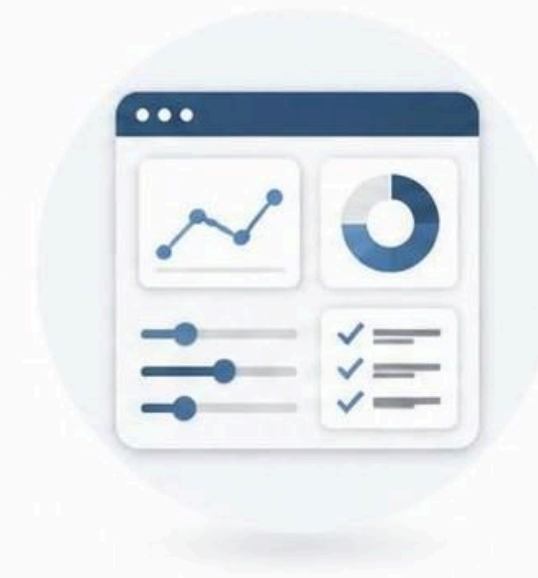
That includes ingestion, storage, processing, access, monitoring, retention, and change management.

Two Questions Regulators Ask



**Is the
record
trustworthy?**

SECURITY • INTEGRITY • TRUST



**Is the
whole
system
controlled?**

GOVERNANCE • OVERSIGHT • CONTROL

Slide 6: Major Aspects

For this talk, I simplify the regulatory discussion into two questions.

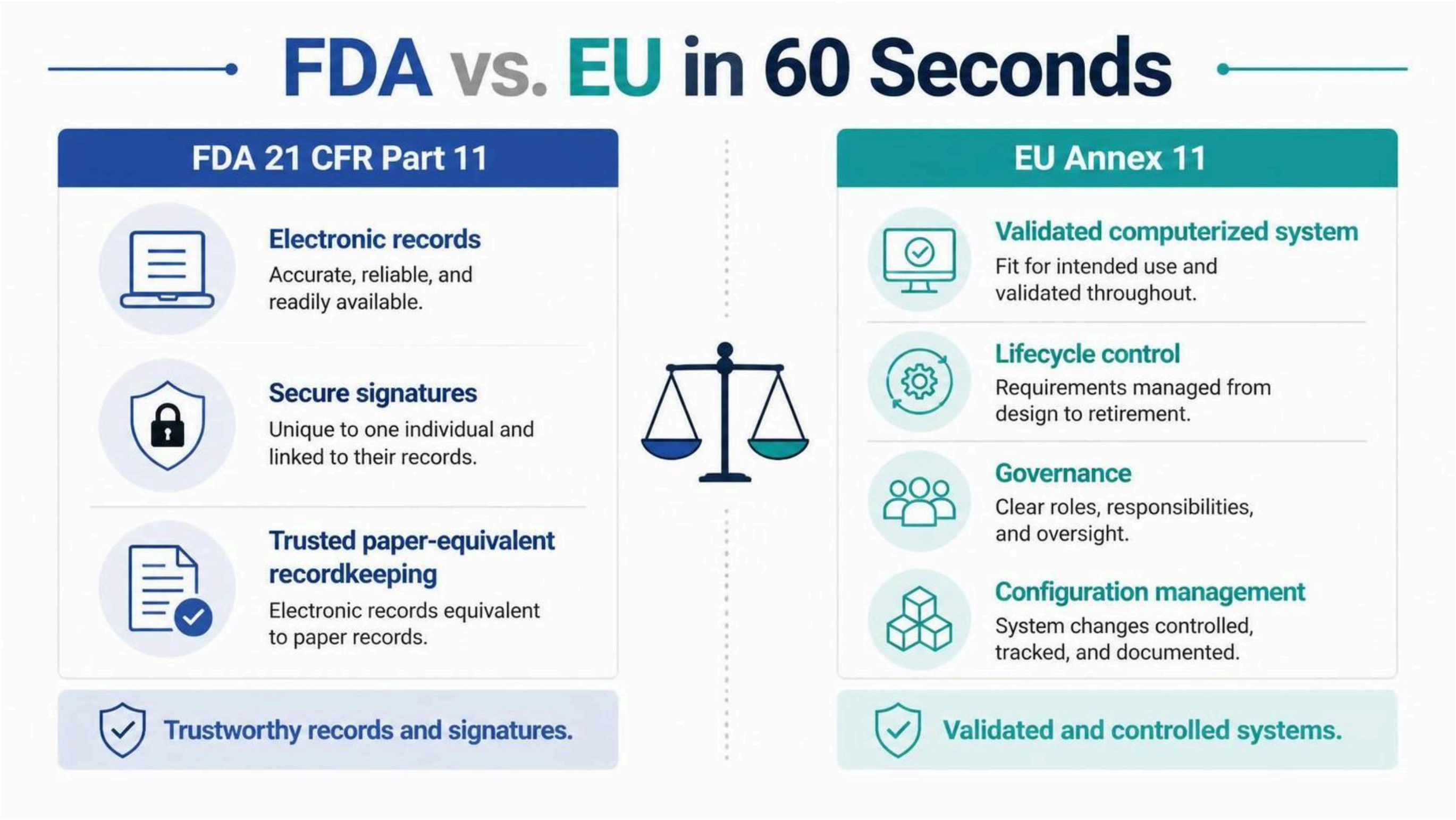
First: is the electronic record trustworthy?

Second: is the whole computerized system controlled?

Everything else in the architecture maps back to these two questions.

The Regulatory Lens





Slide 8: Part 11 vs. Annex 11 in 60 Seconds

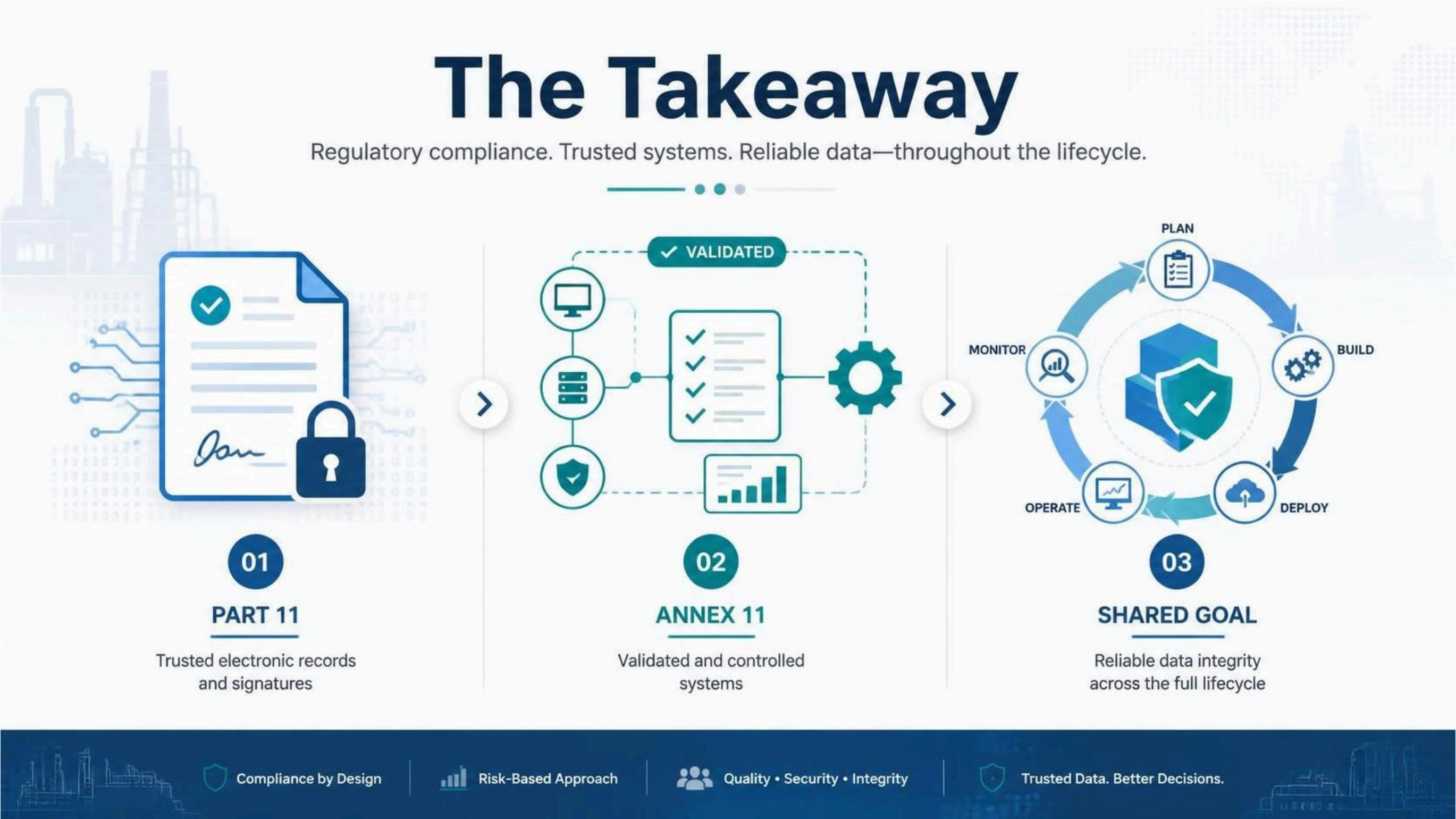
Those two questions map nicely to two regulatory perspectives.

FDA 21 CFR Part 11 focuses on whether electronic records and electronic signatures can be trusted and treated as equivalent to paper-based records.

EU GMP Annex 11 takes a broader lifecycle view. It asks whether the computerized system is validated, governed, secured, monitored, and controlled throughout its lifecycle.

The simplified view is: Part 11 focuses on trusted records and signatures. Annex 11 focuses on validated and controlled systems.

The Shared Compliance Goal



Slide 9: The Shared Compliance Goal

The takeaway is simple.

Part 11 focuses attention on trusted electronic records and signatures.

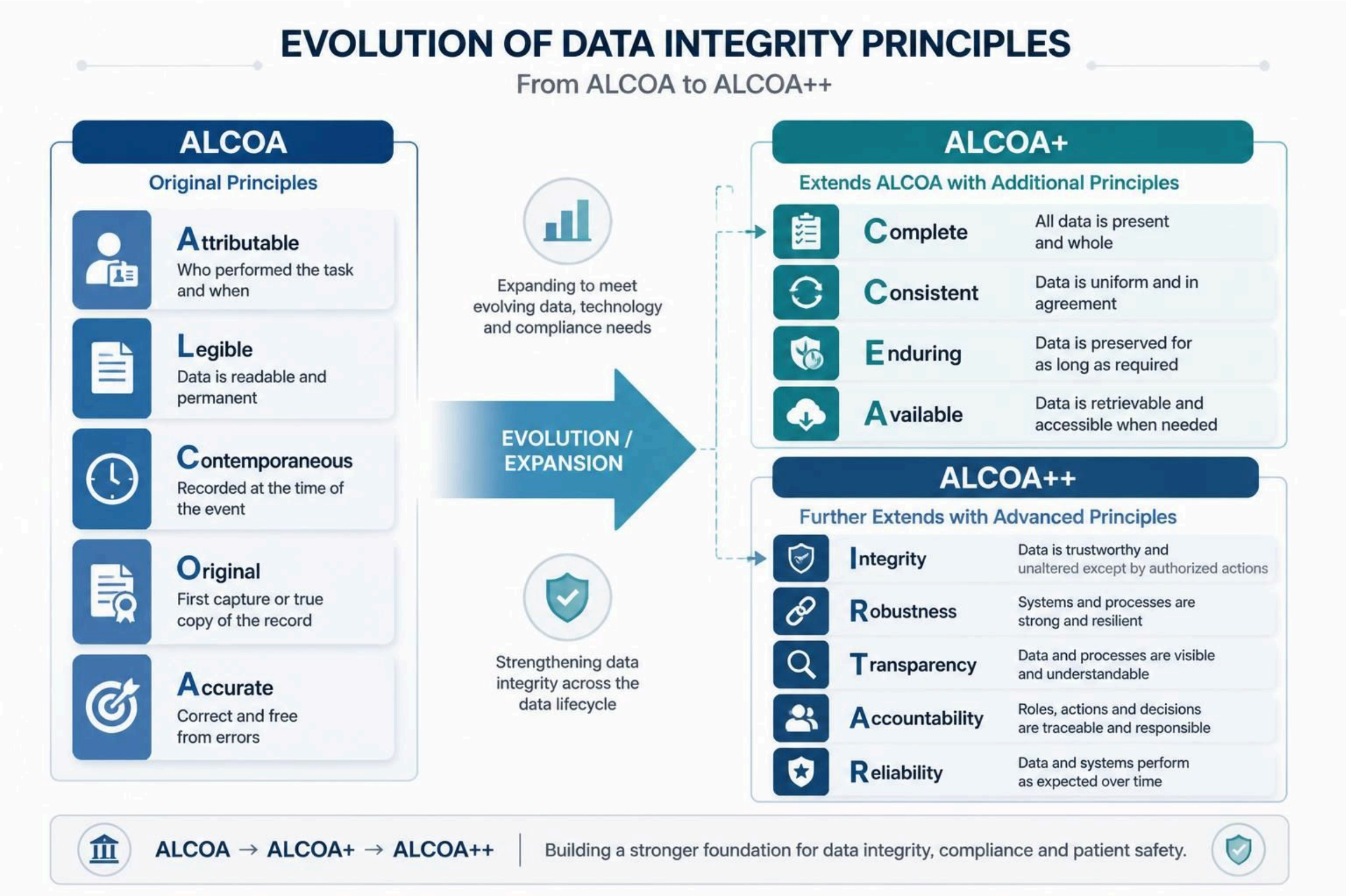
Annex 11 expands the view to the computerized system lifecycle.

Together, they push us toward one shared goal: reliable data integrity from capture to processing, storage, retention, and inspection.

Data Integrity Becomes Architecture



ALCOA++ as the Practical Lens



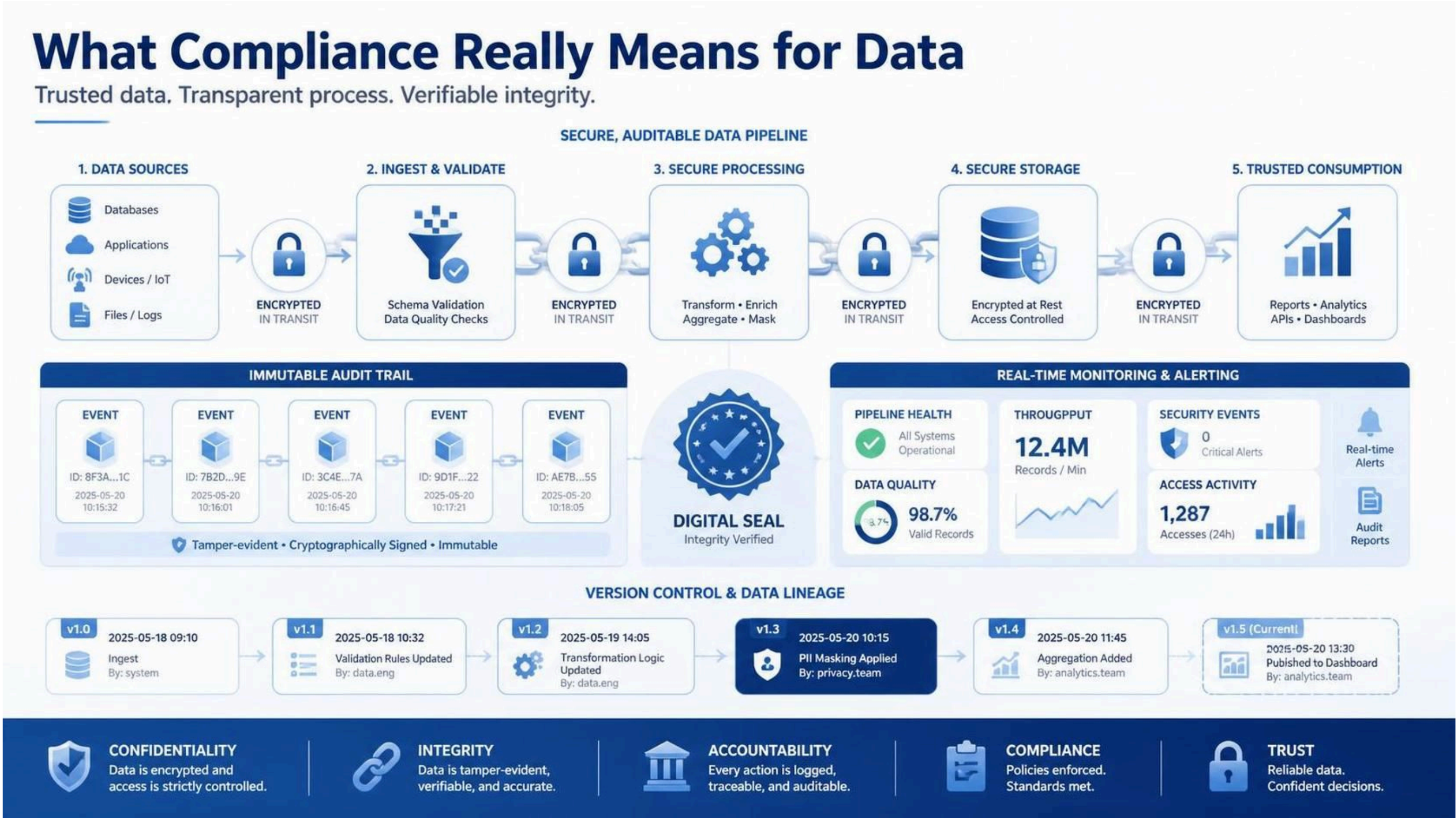
Slide 11: ALCOA++ as the Practical Lens

ALCOA is often introduced as a data integrity framework: attributable, legible, contemporaneous, original, and accurate.

ALCOA+ extends that with complete, consistent, enduring, and available.

For IIoT, we need to translate those principles into architecture. This is no longer only about a human entering data into a form. It is about machine-generated data, automated ingestion, buffering, synchronization, transformation, storage, and long-term availability.

I do not want to walk through every term on this slide. The important point is the evolution: data integrity becomes an end-to-end platform requirement.



Slide 12: Compliancy and Data

This slide translates compliance into a data pipeline.

It is not only encrypted storage. It is a controlled data journey: validated ingestion, protected processing, secure storage, traceable consumption, audit trails, lineage, monitoring, and alerting.

At every stage we need to answer: where did the data come from, what happened to it, who or what accessed it, and can we prove that later?

What Can Go Wrong on the Shopfloor?

Network outage during production	Duplicate readings after retry
Missing or inconsistent timestamps	Unauthorized data access
Silent data loss during synchronization	Uncontrolled transformation logic
Retention or backup failure	Configuration drift between environments

A GMP-ready IIoT architecture must be designed for failure, not only for normal operation.

Slide 13: What Can Go Wrong on the Shopfloor?

This is important because real manufacturing environments are not perfect.

Networks fail. Gateways restart. Messages are retried. Timestamps can differ. Configuration can drift. Data can be transformed without enough traceability.

A credible IIoT data architecture must handle these failure modes in a controlled and observable way.

Why Open Source?



Why Open Source Helps Validation



Slide 15: Why Open Source Helps Validation

Open source does not remove validation work. It does not make a system compliant by default.

But open source gives us something valuable in regulated environments: transparency, inspectable architecture, open interfaces, community review, and less dependency on a single vendor roadmap.

For regulated industries, that matters because validation depends on understanding and controlling the implemented system.

Open Source Is Not Automatically Compliant

- The software is not validated by default.
- The implemented system must be qualified and validated.
- Supplier and community governance still need assessment.
- Security patching and dependency management must be controlled.
- Documentation, testing, SOPs, and change control remain mandatory.

Open source is not a shortcut around GMP. It is a transparent foundation that can support validated architectures.

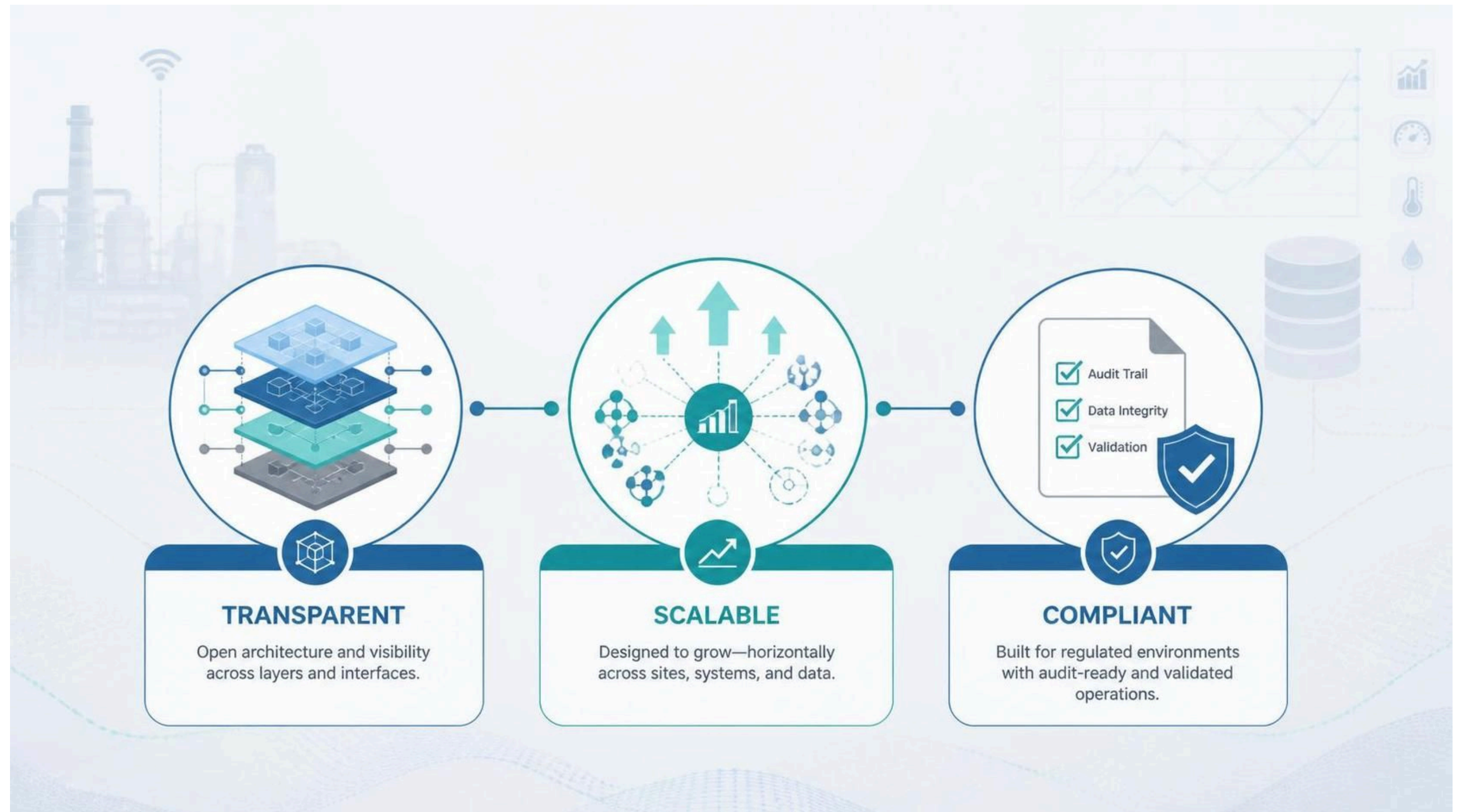
Slide 16: Open Source Is Not Automatically Compliant

This is an important distinction.

We should not say that an open source component is automatically compliant. Compliance comes from the implemented and validated system.

But transparency, open standards, and inspectable behavior can make technical assessment, integration, and long-term governance easier.

What to Remember About Open Source



Slide 17: What to Remember About Open Source

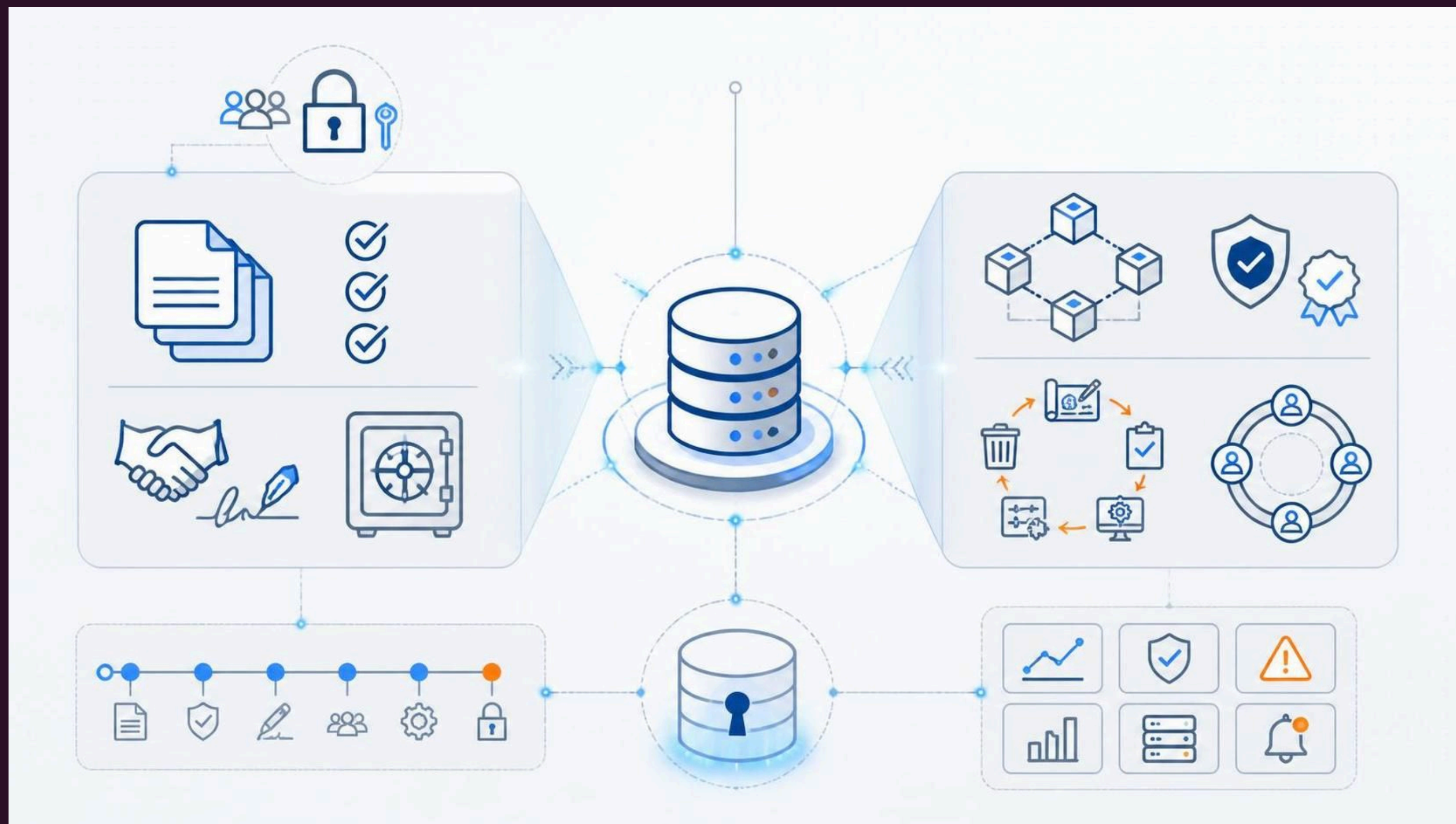
If there are three things to remember about open source in this context, they are transparency, scalability, and validatability.

Transparency means open interfaces and inspectable architecture.

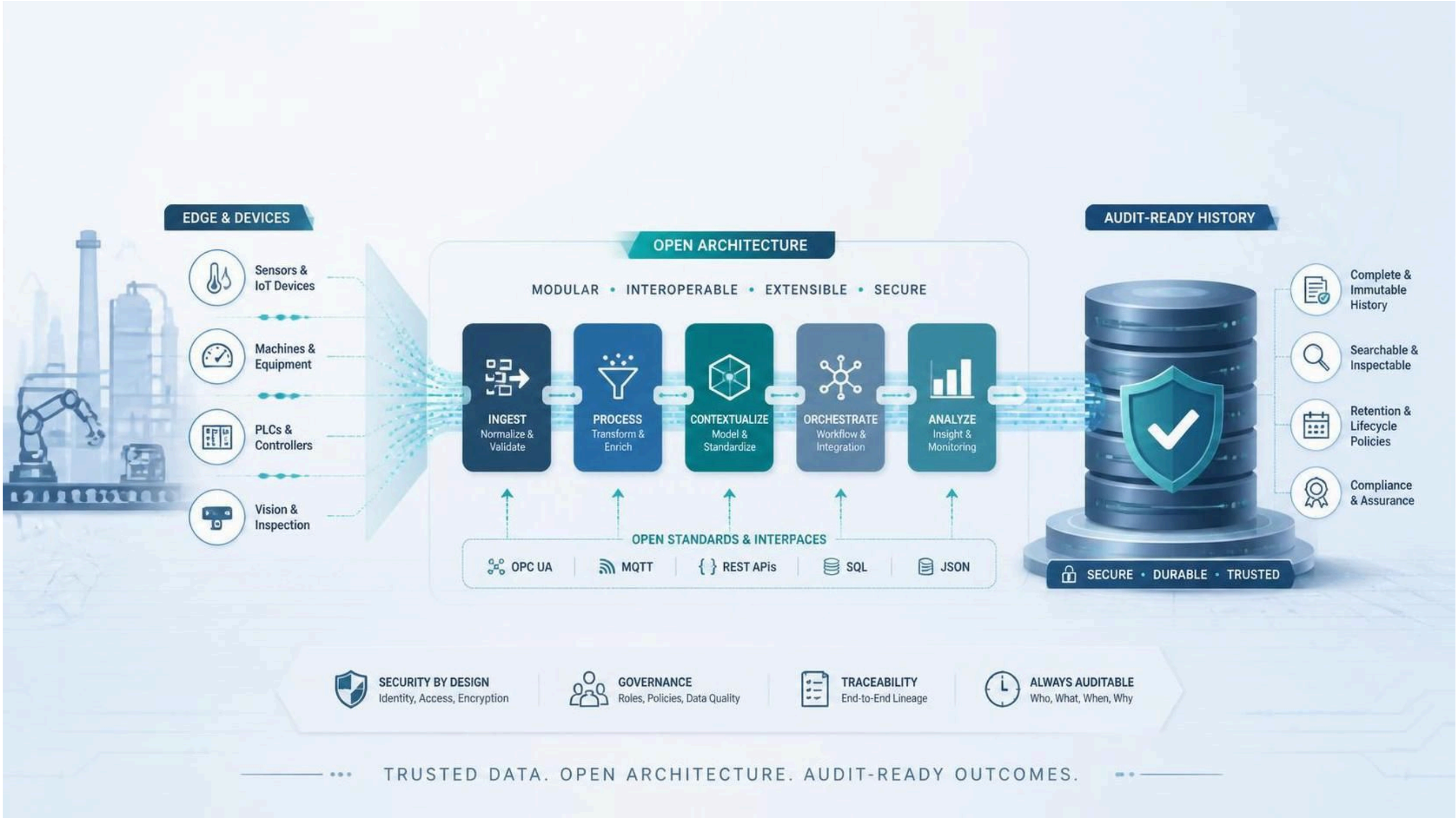
Scalability means the architecture can grow across machines, sites, and data volumes.

Validatability means the system can be specified, tested, monitored, documented, and controlled.

From Open Architecture to Apache IoTDB



Open Architecture for Audit-Ready History



Slide 19: Open Architecture for Audit-Ready History

Before we look at individual technologies, this is the architecture pattern.

We start with edge devices, machines, PLCs, sensors, and inspection systems.

We ingest and validate signals. We process and contextualize data. We orchestrate workflows. We analyze and visualize information.

The outcome must be audit-ready history: searchable, inspectable, retained, and governed.

The bottom layer is critical: security, governance, traceability, and auditability are not add-ons. They must run across the full architecture.

Example Intended Use

- Capture high-frequency machine and sensor readings.
- Buffer data locally when central connectivity is unavailable.
- Synchronize data into a historian without duplicates.
- Visualize current and historical data.
- Support traceability, operational resilience, and inspection readiness.

The goal is not to validate a database in isolation. The goal is to validate the implemented system for its intended use.

Slide 20: Example Intended Use

In regulated environments, intended use is central.

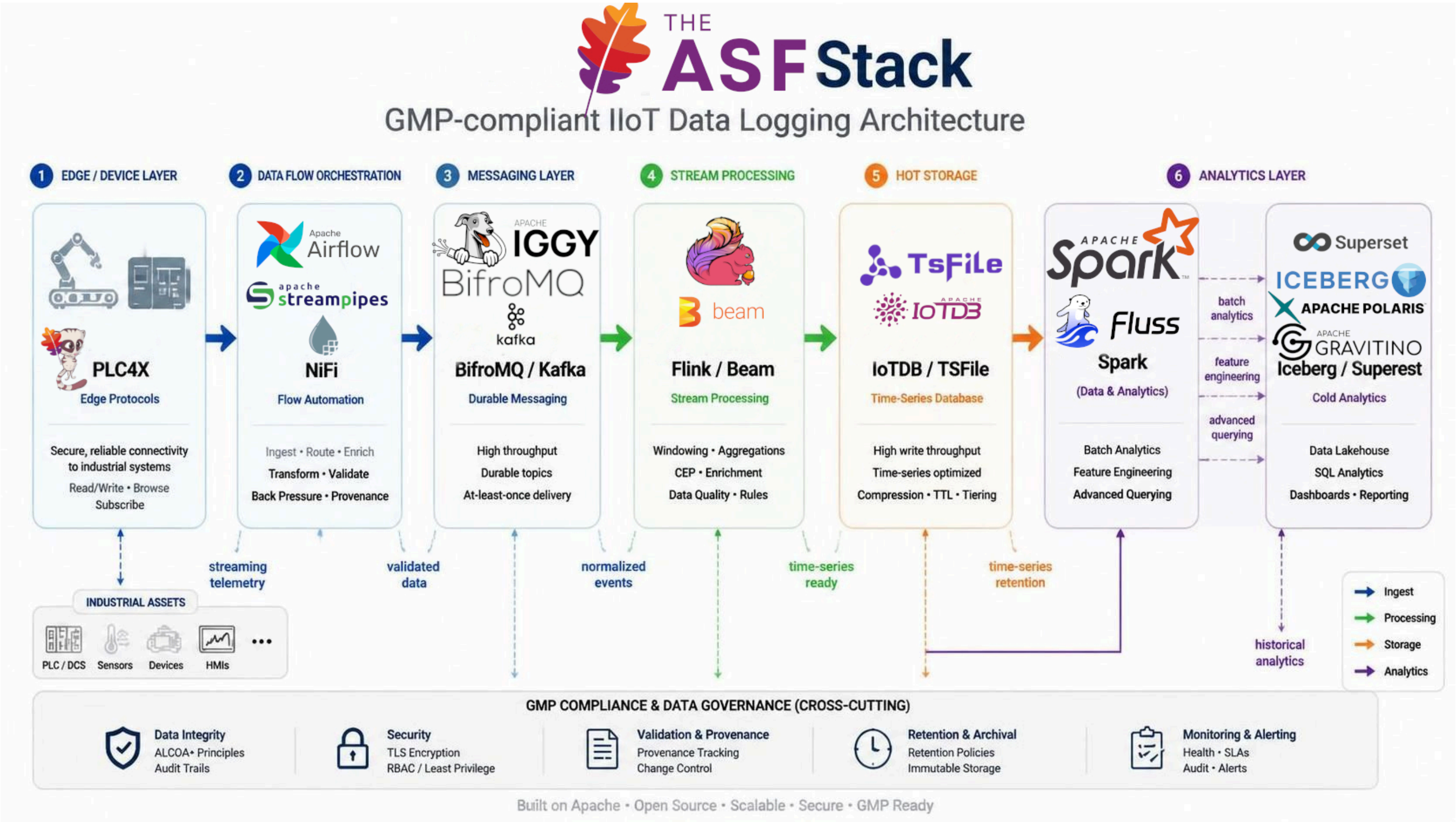
For this example, the intended use is to capture, buffer, store, synchronize, and visualize high-frequency machine data from regulated manufacturing equipment.

The system supports operational monitoring, data integrity, and inspection readiness.

It does not replace the full quality management system, electronic batch record approval, or final product release process.

Apache-Based IIoT Data Logging

Reference Architecture



Slide 21: Apache-Based IIoT Data Logging +

This slide is not saying every project is mandatory. It shows a modular Apache-based reference architecture.

For this talk, the core is Apache IoTDB and TSFile as the historian foundation.

Apache PLC4X can help connect to industrial protocols.

Apache NiFi can orchestrate controlled data flows.

Kafka or BifroMQ can provide buffering and decoupling.

Flink or Beam can process streams.

Iceberg, Spark, and Superset can support analytical consumption and visualization.

The most important part is the cross-cutting layer: data integrity, security, validation evidence, retention, monitoring, and alerting.

Apache IoTDB as the Historian Core

- Optimized for high-frequency industrial time-series data.
- Stores structured device and sensor measurements.
- Supports integration with authentication, authorization, encryption, monitoring, and retention controls.
- Can serve as the historian core inside a validated IIoT architecture.
- Works together with controlled ingestion, metadata, logging, and validation processes.

Slide 22: Apache IoTDB as the Historian Core

Apache IoTDB is the time-series database at the center of the architecture.

The important wording here is that IoTDB can serve as the historian core inside a validated architecture.

The database alone does not make the system compliant. The system becomes GMP-ready when the architecture around it provides controlled ingestion, traceability, access control, retention, monitoring, testing, and lifecycle governance.

Apache TSFile as the Storage Foundation

- Time-series file format optimized for efficient storage and query.
- Supports append-oriented storage patterns.
- Helps separate raw time-series history from downstream analytics.
- Supports reproducibility and inspection when combined with controlled storage, access, and retention design.
- Can be paired with analytical formats such as Apache Iceberg.

Slide 23: Apache TSFile as the Storage Foundation

TSFile is the underlying time-series file format used by IoTDB.

Append-oriented storage patterns are useful because they support preservation of historical data and reproducibility.

However, immutability is not only a file-format property. A tamper-evident and inspection-ready design also depends on access control, storage policies, backups, checksums, monitoring, and operational procedures.

The Demo: Resilience and Traceability



Demo Scenario

Network Outage Without Data Loss

- A production line continues generating readings.
- The central historian is temporarily unavailable.
- The local system continues accepting data.
- Readings are buffered safely.
- Synchronization resumes when connectivity returns.
- Duplicate records are avoided through idempotent behavior.
- Historical data becomes visible in Apache IoTDB.

The demo proves continuity, traceability, and recoverability under realistic shopfloor conditions.

Slide 25: Demo Scenario +

The demo focuses on one realistic shopfloor problem: connectivity is not perfect.

A production line may continue generating readings even when the central historian is temporarily unavailable.

The system must keep accepting data locally, buffer it safely, and synchronize later without duplicates or silent data loss.

That is the pattern I want to show now.

Live Demo: Hybrid IoT Data Logging and Visualization

- Goal: show the path from ingestion to buffering to synchronization.

Slide 26: Live Demo: Hybrid IoT Data Logging and Visualization

During the demo, focus on the control points.

First, authenticated ingestion.

Second, local buffering.

Third, synchronization progress.

Fourth, historical visibility in IoTDB.

Fifth, idempotent resume behavior, meaning that repeating the operation does not create duplicates or incorrect state.

Demo Steps

Step	Action
1	Log in as <code>operator</code> with <code>operator</code> .
2	Send sample readings to <code>/ingest</code> or use the ingestion helper.
3	Open the dashboard and observe the local buffer chart.
4	Trigger <code>Sync to IoTDB</code> and watch the live SSE progress stream.
5	Verify that synchronized historical data appears in the IoTDB chart.

Slide 27: Demo Steps

If time is short, run only ingestion and synchronization, then point to the dual-chart comparison and recovery behavior.

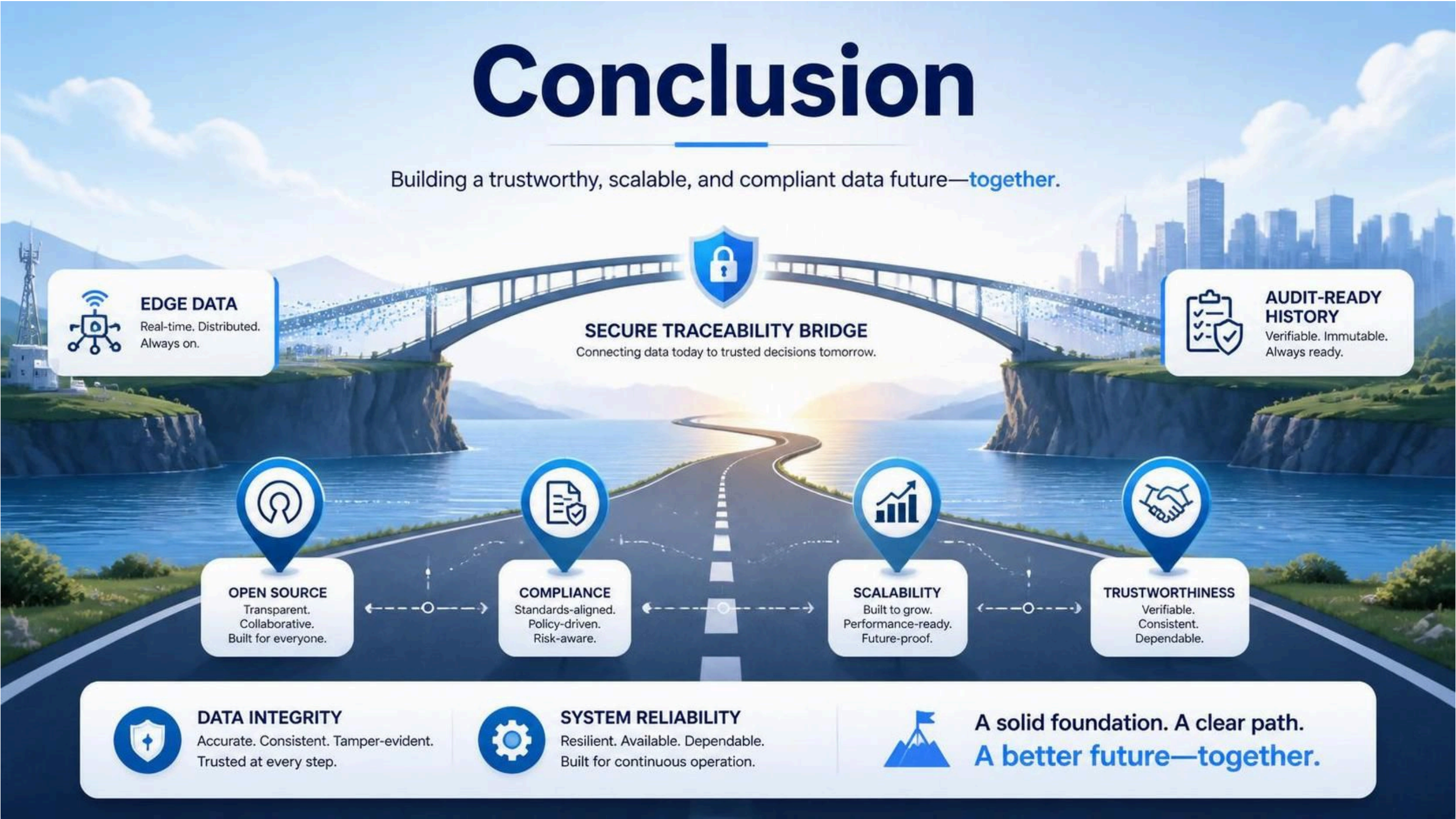
SSE means Server-Sent Events.

Idempotent means repeating the action does not create duplicates or incorrect state.

Conclusion



From Edge Data to Audit-Ready History



Slide 29: From Edge Data to Audit-Ready History

The future of regulated manufacturing will be more connected, more automated, and more data-driven.

But the core questions remain simple.

Can we trust the record?

Can we prove the system is controlled?

Apache IoTDB and the Apache ecosystem give us a strong open foundation for high-frequency IIoT data logging.

But compliance is created by the full validated system: architecture, controls, evidence, monitoring, procedures, and change control.

Open source is not a shortcut around GMP. It is a transparent foundation we can validate and evolve together.

Final Takeaways

- Regulated manufacturing does not just need more data.
- It needs data that can be trusted, inspected, and preserved.
- Apache IoTDB can serve as the historian core of a GMP-ready IIoT architecture.
- Apache technologies provide modular building blocks from edge to analytics.
- Validation depends on intended use, risk, evidence, procedures, and lifecycle control.
- Open source supports transparency, scalability, and validatability.

Open source is not a shortcut around validation. It is a foundation for transparent, scalable, and inspectable manufacturing data platforms.

Slide 30: Final Takeaways

If there is one message to remember, it is this:

Apache IoTDB can serve as the historian core of a GMP-ready IIoT architecture, but compliance does not come from the database alone.

It comes from the validated system around it: controlled ingestion, traceability, access control, retention, monitoring, testing, and lifecycle governance.

Check out:

- Git Repositories
 - https://github.com/ottlukas/iot-data-logging-hybrid_2026
 - https://github.com/ottlukas/coc_iiot_talk_2026
 - https://github.com/ottlukas/ApacheCon_2022_IoT
- Projects:
 - Apache IoTDB: <https://iotdb.apache.org/>
 - Apache PLC4X: <https://plc4x.apache.org/>
 - Apache StreamPipes: <https://streampipes.apache.org/>
- Contact:
 - Lukas Ott <otluk@apache.org>

Slide 31: Check out:

Thank you for your attention. I'm happy to take questions.